



Evaluation of London House Prices a Case Study

GIS AND STATISTICAL MODELLING TASK - 1

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CONTENTS

INTRODUCTION	3
Spatial basis of the problem	3
METHOD	3
Data sources and Study Population.....	3
Study Area	4
Study Variables	4
RESULTS	4
Descriptive Statistics	4
Spatial Distribution of House Prices in London	5
POTENTIAL THEORIES	6
Spatial Distribution of Jobs and House Prices in London.....	6
Spatial Distribution Median House Price and Population density	7
Spatial Distribution Median House Price and New Houses	8
CONCLUSION	9
REFERENCES	10

INTRODUCTION

The price of a house is typically determined by the interaction between the demand and supply in the market. However, there are situations where the supply of houses doesn't increase even when the price rises, or the price is driven by factors other than the basic need for housing and ability to pay, leading to high housing costs that are unaffordable for many. This can have negative effects on the economy, including higher wages and business costs, and a greater risk of instability. In economic terms, housing affordability is defined as the ability to pay for housing in relation to the market-determined housing costs. This definition does not include "affordable homes," which refers to housing provided to eligible households who are unable to find suitable housing in the market. (Marsden, 2015).

In the past few decades, there has been a significant amount of research exploring the characteristics and dynamics of housing markets. This is not unexpected given the numerous reasons that demonstrate the significance of housing, such as: the significant contribution of housing to an individual's net worth; the role of housing in the overall economy; the way housing interacts with economic fundamentals; and the growing importance of housing, compared to the stock market, in influencing consumption decisions. (Webb et al., 2019).

This paper examines the issue of housing affordability in London and the potential risks of instability in the housing market. The focus is on the economic perspective, analyzing the factors that drive the market for buying and selling homes. The paper discusses recent trends in the housing market and reviews the data sources used for analysis, along with the limitations of the data. The paper analyzes whether house prices in London are reasonable based on economic fundamentals, using a range of indicators.

Spatial basis of the problem

The demand for housing in London is influenced by multiple factors including the appeal of living and working in the city, as well as natural population changes and socio-economic factors that lead to more households. This demand, combined with the greater affordability of buying homes, causes housing prices in London to rise. A properly functioning housing market should respond to this demand by increasing the supply of housing available. (Bissoondeal & Tsiaras, 2021).

In view of these evidences, this report aims to:

- Describe spatial distribution of house prices in London.
- Examine existence of any spatial pattern's indicative of spatial autocorrelation.
- Advise probable theory in relation to spatial patterns spotted.

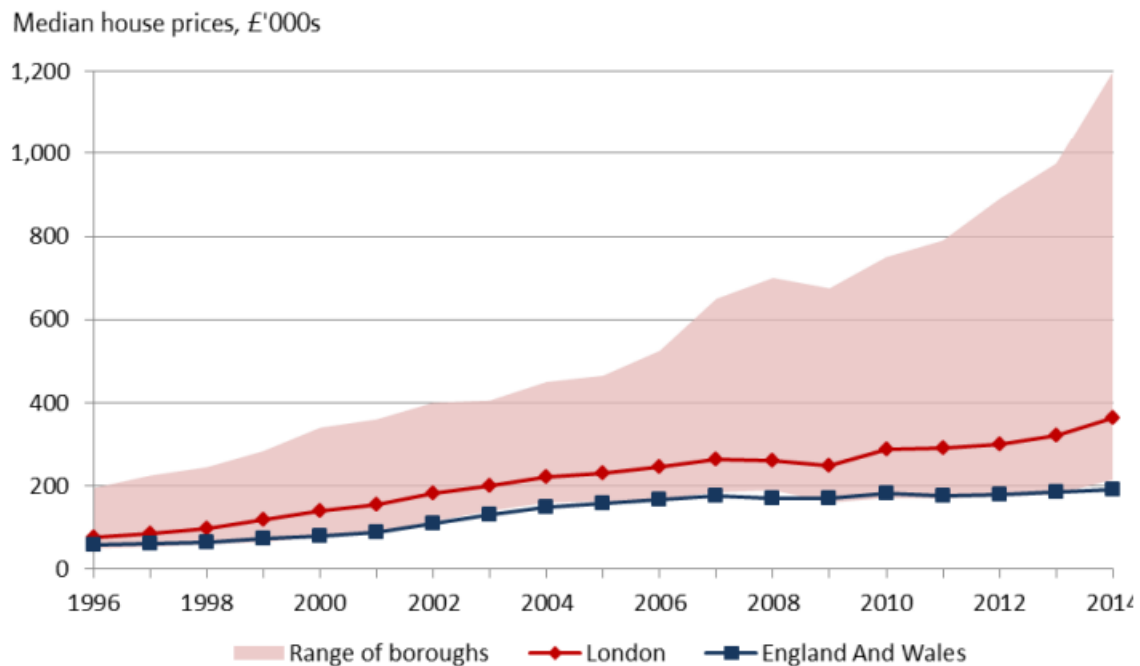
METHOD

Data sources and Study Population

The ward level estimated house price data from Urban Big Data Center (UBDC) for the year 2013 has been used for this report (London Borough Profiles - UBDC Data Portal (n.d.)).

Study Area

To determine if there are any spatial patterns that suggest spatial autocorrelation in the local authority districts of London, this report focuses on this geographical study area. London was selected because it has the highest house prices, making it a relevant location to examine in terms of the housing market.



Source: Land Registry price paid data. Notes: Excluded from the above figures are sales at less than market price (e.g. Right to Buy), sales below £1,000 and sales above £20 million.

Study Variables

The dependent variables used in this analysis are job population and new homes. These variables are expected to have an impact on the housing market in the local authority districts of London. Job population can influence demand for housing, while new homes can affect the supply of available housing. By examining the relationship between these variables and the housing market, insights can be gained into potential drivers of housing prices and overall market trends.

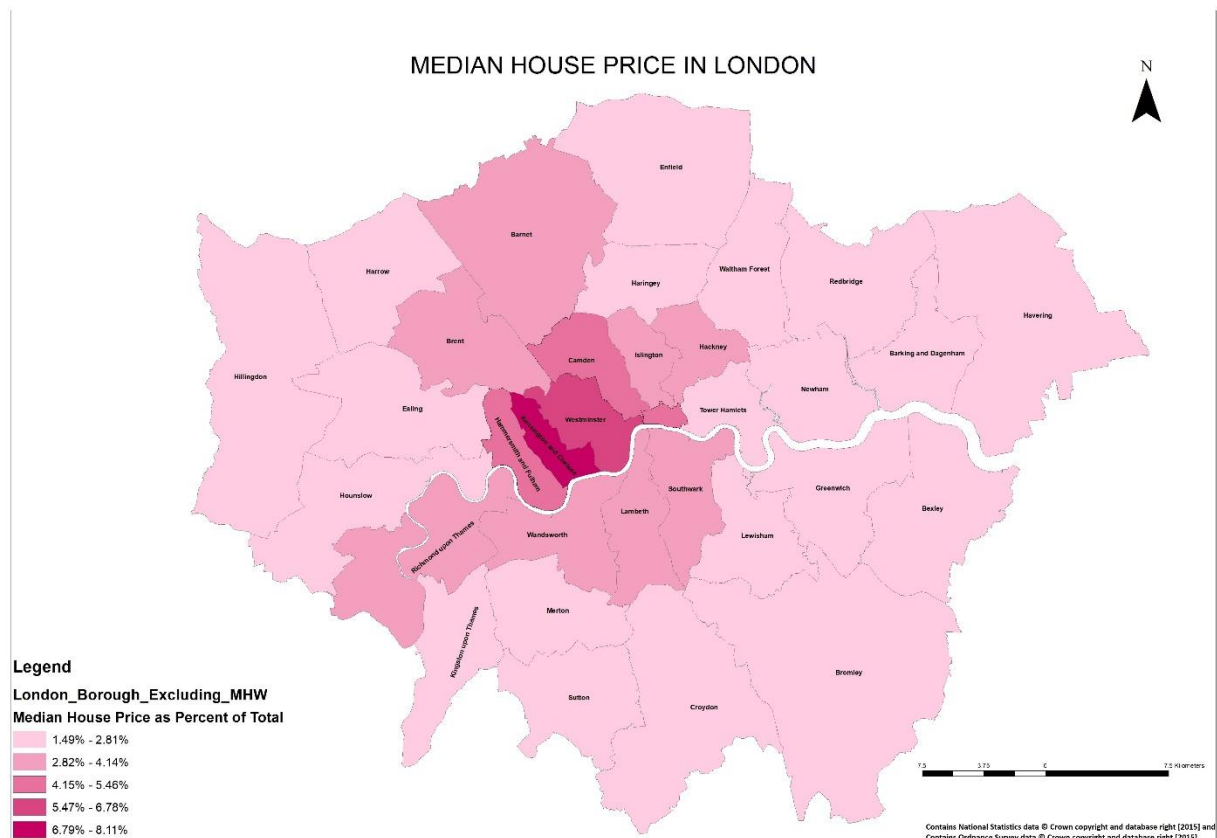
RESULTS

Descriptive Statistics

Table 1 : Pervasiveness of House prices in London for year 2014

Count:	Minimum:	Maximum:	Sum:	Mean:	Standard Deviation:	Nulls:
33	180000	980000	12088875	366329.5	163782.7	0

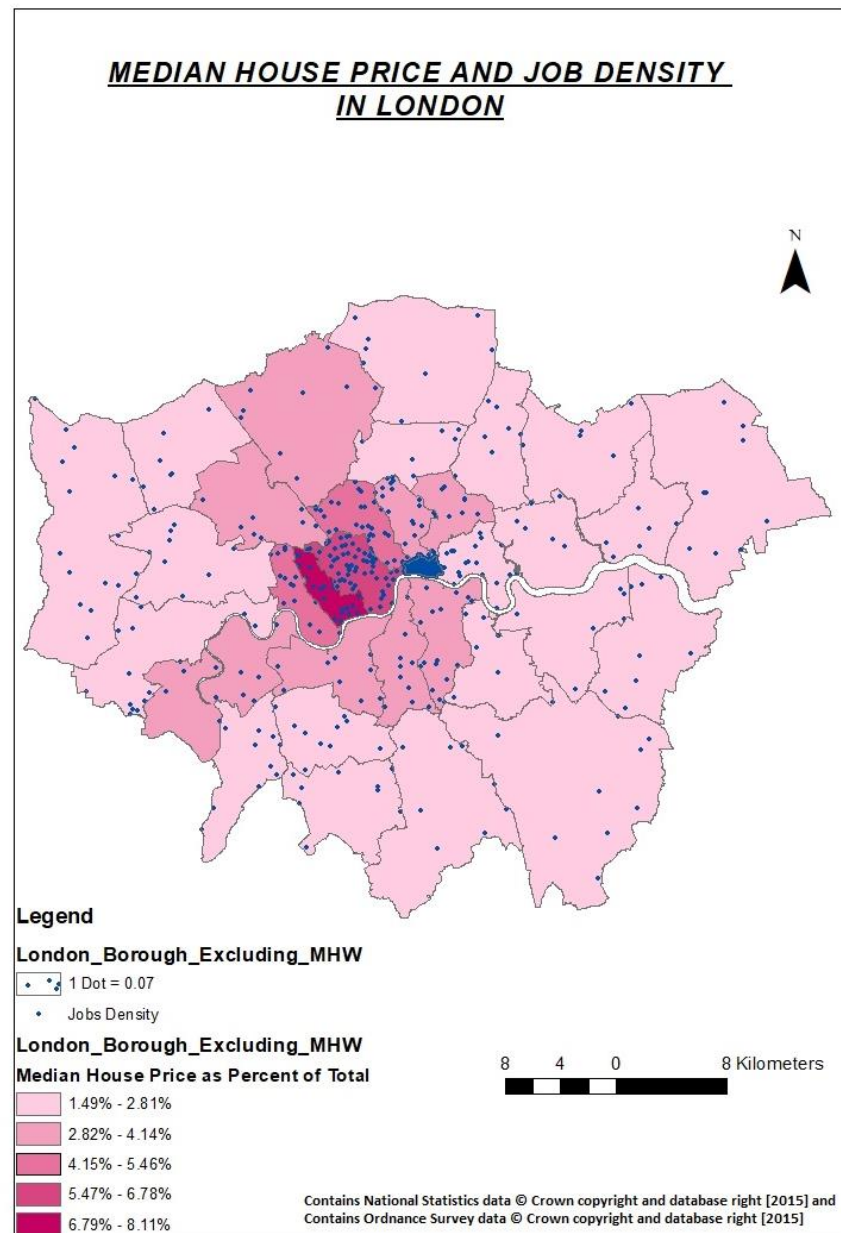
Spatial Distribution of House Prices in London



The most affluent boroughs, grouped as the Top Tier, are geographically concentrated in the central part of London, while the least affluent boroughs, known as the Lowest Tier, are dispersed throughout the outskirts of Greater London. The Middle Tier boroughs are primarily located in the northwest region of London, with a few exceptions in the southwest and east. However, all the Middle Tier boroughs share a border with at least one borough from the Top Tier group. (Tsai & Wang, 2021).

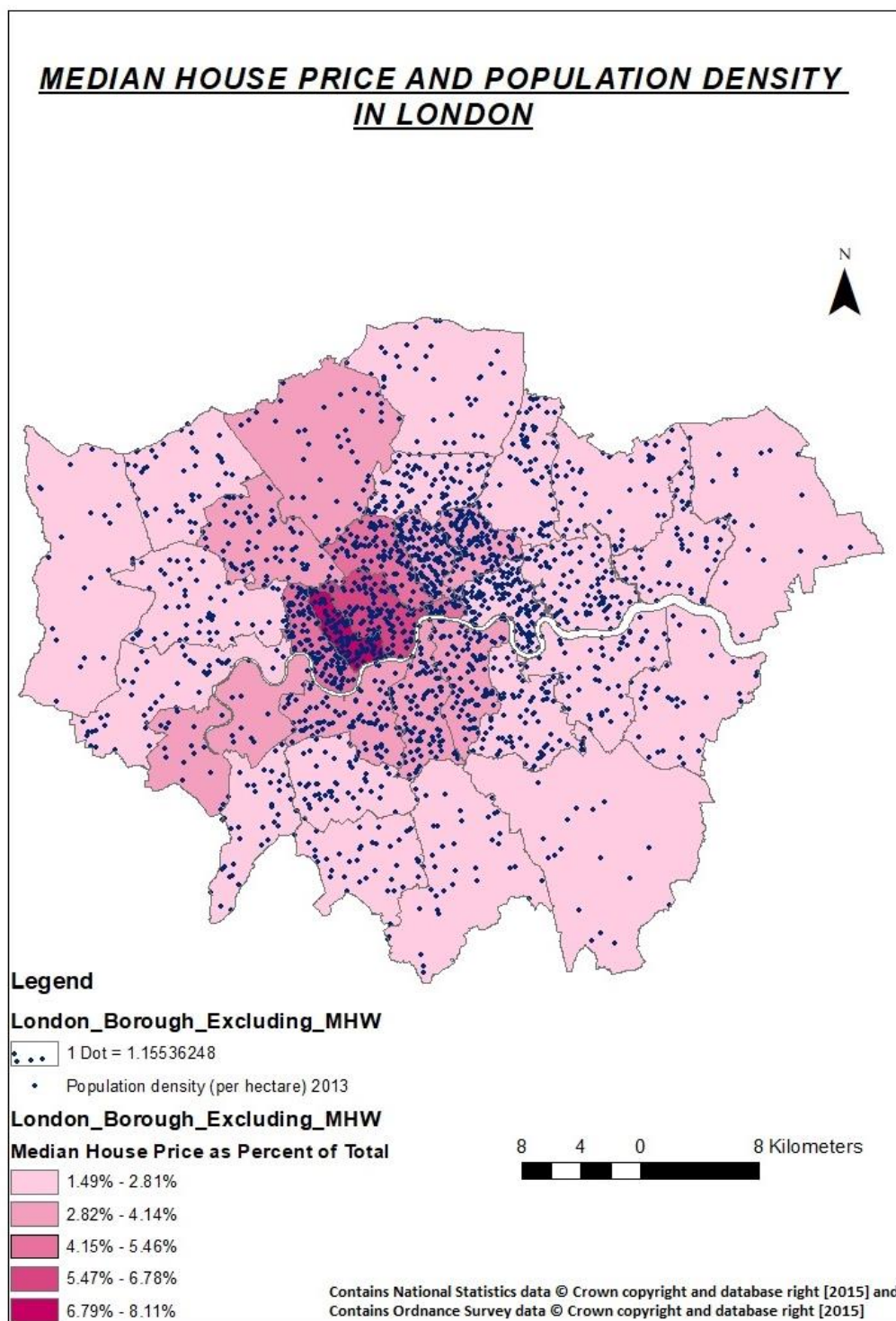
POTENTIAL THEORIES

Spatial Distribution of Jobs and House Prices in London



The study revealed that the proximity to London had a positive correlation with property prices, but only up to a certain distance. For, towns located less than 200 kilometers away from London, the data showed a decrease of £50,000 in property price for every extra 50 kilometers of distance from the city. However, it's important to note that the limit could represent a commutable distance, and travel time, distance travelled, and cost may not be directly related to direct route distance. Additionally, the study did not find strong indication of a comparable effect from other urban zones, but more research is needed to confirm this. (Office for National Statistics, 2019 n.d.)

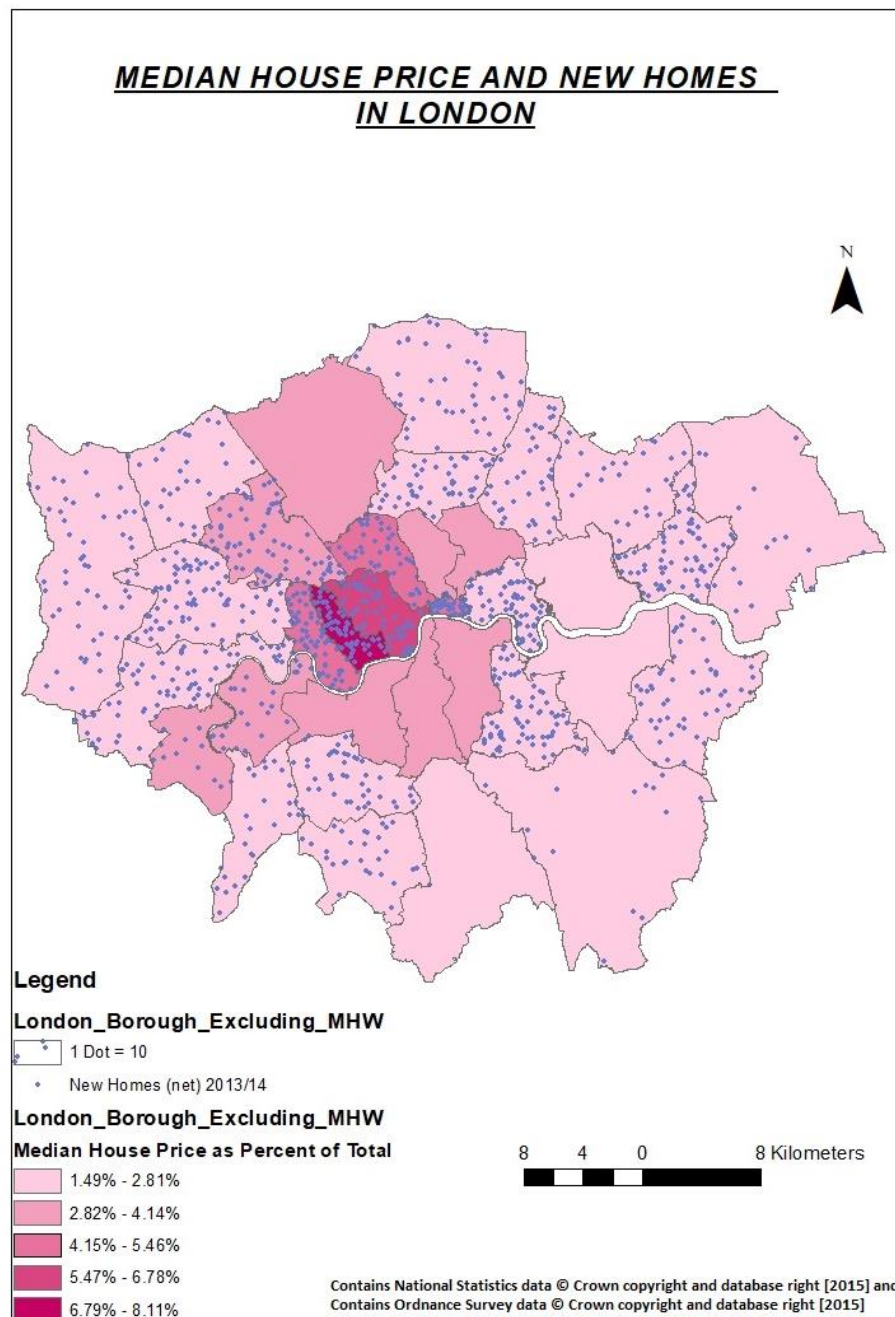
Spatial Distribution Median House Price and Population density



Areas with higher population density tended to have higher house prices, and this was especially true for central London. This was likely due to factors such as increased demand for housing in more densely populated areas, as well as the limited availability of land for new construction in central London. However, some areas with high population density, such as those in the east and southeast of London, did not necessarily have high house prices, which

suggests that other factors also played a role in determining property values. Overall, the relationship between population density and house prices in London was complex and multifaceted, and was influenced by a range of economic, social, and geographical factors.

Spatial Distribution Median House Price and New Houses



In London, the supply of new housing tends to be relatively unresponsive to market signals of increased demand, even when house prices rise. This lack of responsiveness is largely due to the fact that the supply of new building tends to be rather inelastic. Additionally, the levels of annual completions have fluctuated from year to year, suggesting that the supply of new housing in London is not keeping pace with demand.

CONCLUSION

The distribution of median house prices in London shows that there are clusters of wards with higher prevalence, which suggests that there may be spatial autocorrelation present in the data.

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